Haesung Kim

Ph.D. in Electrical Engineering

Postdoctoral Researcher

Specere Lab, School of Mechanical Engineering, Birck Nanotechnology Center, Purdue University, West Lafayette, IN

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Research Focus

Opto-electronics-based characterization of semiconductor devices, physics-based device modeling, and the development of parameter-extraction methods. The work centers on extracting device parameters that are otherwise difficult to separate — parasitic resistances, interface and subgap trap distributions, and the conduction band minimum — using opto-electronic and transient measurements, applied across silicon MOSFETs (including gate-all-around and NAND flash structures), amorphous IGZO thin-film transistors, ferroelectric FETs, and ferroelectric TFTs.

Education

2022.03 – 2025.08	**Ph.D. in Electrical Engineering** Kookmin University, Seoul, Korea Advisers: Prof. Dong Myong Kim, Prof. Jong-Ho Bae
2020.03 – 2022.02	**M.S. in Electrical Engineering** Kookmin University, Seoul, Korea
2014.03 – 2020.02	**B.S. in Electrical Engineering** Kookmin University, Seoul, Korea

Appointments

2025.08 – present	**Postdoctoral Researcher** Specere Lab, School of Mechanical Engineering Birck Nanotechnology Center, Purdue University, West Lafayette, IN, USA Advisor: Prof. Thomas Beechem
2025.03 – 2025.06	**Lecturer** Kookmin University, Seoul, Korea Intelligent Semiconductor Devices (3 credits, junior level)
2015.09 – 2017.06	**Republic of Korea Marine Corps** Mandatory military service

Journal Articles

1. Y. Choi, H. Yang, H. J. Kim, S. Park, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, **H. Kim**, J. H. Bae, "Characterization of trap dynamics via transient response in amorphous InGaZnO thin-film transistors," *Applied Physics Letters*, vol. 128, no. 24, 2026. doi:10.1063/5.0332203

2. H. J. Kim, H. Yang, **H. Kim**, S. Park, H. N. Lee, S. J. Choi, D. M. Kim, D. H. Kim, M. K. Park, J. H. Bae, "Analysis of spatial distribution of remnant polarization in inverted staggered a-IGZO/HZO ferroelectric thin-film transistor using capacitance-voltage characteristics," *Solid-State Electronics*, vol. 235, pp. 109358, 2026. doi:10.1016/j.sse.2026.109358

3. **H. Kim**, J. Y. Park, S. H. Han, H. Yang, Y. J. Lee, S. J. Choi, D. H. Kim, J. H. Bae, D. M. Kim, "Characterization of Low-Temperature Intrinsic Field-Effect Mobility With Contact Resistances in Amorphous InGaZnO Thin Film Transistors," *IEEE Transactions on Electron Devices*, vol. 73, no. 7, pp. 4341–4345, 2026. doi:10.1109/TED.2026.3691120

4. **H. Kim**, J. Park, J. Im, H. Kim, Y. M. Kim, K. Park, S. J. Choi, D. H. Kim, D. M. Kim, Y. J. Yoon, S. Y. Woo, J. H. Bae, "Impact of proton irradiation on SiNx and Si–SiO2 interfaces in FLASH memory cell," *Nuclear Engineering and Technology*, vol. 58, no. 10, pp. 104433, 2026. doi:10.1016/j.net.2026.104433

5. Y. Kim, Y. J. Lee, J. Yang, B. Kim, S. J. Kim, J. Kim, I. Im, J. Y. Kim, H. Rui, **H. Kim**, C. W. Bark, M. H. Park, D. H. Kim, S. J. Choi, J. J. Yang, S. Lee, H. W. Jang, "Programmable ferroelectric rectifier for reliable and efficient neuromorphic crossbar array," *Nature Communications*, vol. 17, no. 1, 2026. doi:10.1038/s41467-026-70727-2
6. S. Kim, H. Yang, H. Lee, J. Park, H. Jeong, S. Y. Woo, M. K. Park, **H. Kim**, J. H. Bae, "Source/drain metal work-function-dependent memory characteristics governed by ferroelectric switching-assisted current modulation in a-IGZO FeFETs," *Materials Science in Semiconductor Processing*, vol. 215, pp. 111021, 2026. doi:10.1016/j.mssp.2026.111021
7. **H. Kim**, H. Yang, Y. Choi, H. Kim, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Unveiling charge trapping dynamics in SiO2/HfZrOX–based ferroelectric field–effect–transistors to minimize read–after–write latency," *Current Applied Physics*, vol. 89, pp. 181–185, 2026. doi:10.1016/j.cap.2026.06.007
8. H. Lee, S. Kim, C. Han, **H. Kim**, H. Yang, S. Park, S. Yun, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Quantitative Analysis on the Interaction Between Channel Carrier and Remote Trap in $Hf_xZr_{1-x}O_2/SiO_2$ Interface in Ferroelectric Field-Effect-Transistor," *Advanced Electronic Materials*, vol. 12, no. 2, 2026. doi:10.1002/aelm.202500549
9. J. Park, **H. Kim**, S. J. Choi, D. H. Kim, D. M. Kim, J. H. Bae, "Exploring low-frequency noise dynamics in a-IGZO TFTs: Unveiling the impact of contact metal variations for advanced semiconductor applications," *Solid-State Electronics*, vol. 231, pp. 109271, 2026. doi:10.1016/j.sse.2025.109271 †
10. H. Yang, **H. Kim**, H. J. Kim, D. M. Kim, S. J. Choi, D. H. Kim, Y. J. Lee, J. H. Bae, "Defect-mediated enhancement of memory window in IGZO-channel ferroelectric field effect transistors," *Materials Science in Semiconductor Processing*, vol. 204, pp. 110268, 2026. doi:10.1016/j.mssp.2025.110268
11. **H. Kim**, S. H. Han, H. Jeong, H. Yang, S. J. Choi, D. H. Kim, J. H. Bae, D. M. Kim, "Photonic I–V Technique With Photogating Effect for Extended Extraction of Subgap Density of States in Amorphous Oxide Semiconductor TFTs," *IEEE Transactions on Electron Devices*, vol. 72, no. 5, pp. 2751–2755, 2025. doi:10.1109/TED.2025.3547707
12. S. Park, H. Yang, **H. Kim**, H. Jeong, S. J. Choi, D. H. Kim, D. M. Kim, M. K. Park, J. H. Bae, "Analysis of operation delay and switching speed limitation due to source/drain resistance and subgap density of states in amorphous InGaZnOx/HfZrOx ferroelectric thin-film transistor," *Solid-State Electronics*, vol. 229, pp. 109203, 2025. doi:10.1016/j.sse.2025.109203
13. S. H. Han, **H. Kim**, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Photovoltaic Effect De-Embedded Photonic C–V Characterization of Subgap Density of States in Amorphous Oxide Semiconductor Thin-Film Transistors," *IEEE Transactions on Electron Devices*, vol. 71, no. 11, pp. 6795–6798, 2024. doi:10.1109/TED.2024.3469161 †
14. **H. Kim**, H. Yang, S. Lee, S. Yun, J. Park, S. Park, H. N. Lee, H. Kim, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Comprehensive analysis of read-after-write latency in HfZrOX-based ferroelectric field-effect-transistors with SiO2 interfacial layer," *Applied Physics Letters*, vol. 124, no. 3, 2024. doi:10.1063/5.0172204
15. S. Lee, **H. Kim**, H. Yang, S. Yun, J. Park, H. Lee, S. Park, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Analysis of the Role of Interfacial Layer in Ferroelectric FET Failure as a Memory Cell," *IEEE Electron Device Letters*, vol. 45, no. 4, pp. 562–565, 2024. doi:10.1109/LED.2024.3360419 †
16. H. Yang, S. Park, S. Yun, **H. Kim**, H. Lee, M. K. Park, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Exploration of structural influences on the ferroelectric switching characteristics of ferroelectric thin-film transistors," *Nanoscale*, vol. 16, no. 42, pp. 19856–19864, 2024. doi:10.1039/d4nr02096k
17. **H. Kim**, H. B. Yoo, H. Lee, J. H. Ryu, J. Y. Park, S. H. Han, H. Yang, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Extraction Technique for the Conduction Band Minimum Energy in Amorphous Indium–Gallium–Zinc–Oxide Thin Film Transistors," *IEEE Transactions on Electron Devices*, vol. 70, no. 6, pp. 3126–3130, 2023. doi:10.1109/TED.2023.3269735 †
18. J. Y. Park, **H. Kim**, H. B. Yoo, J. H. Ryu, S. H. Han, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Simultaneous Extraction of Mobility Enhancement Factor and Threshold Voltage With Parasitic Resistances in Amorphous Oxide Semiconductor Thin-Film Transistors," *IEEE Transactions on Electron Devices*, vol. 70, no. 5, pp. 2312–2316, 2023. doi:10.1109/TED.2023.3253468 †
19. **H. Kim**, H. B. Yoo, J. H. Ryu, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Current-to-transconductance ratio technique for simultaneous extraction of threshold voltage and parasitic resistances in MOSFETs," *Solid-State Electronics*, vol. 183, pp. 108133, 2021. doi:10.1016/j.sse.2021.108133 †

20. H. B. Yoo, **H. Kim**, J. H. Ryu, J. Park, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Modeling and characterization of photovoltaic and photoconductive effects in insulated-gate field effect transistors under optical excitation," *Solid-State Electronics*, vol. 186, pp. 108139, 2021. doi:10.1016/j.sse.2021.108139

21. J. Yu, H. B. Yoo, **H. S. Kim**, J. H. Ryu, S. J. Choi, D. H. Kim, D. M. Kim, "Characterization of Spatial Distribution of Trap Across the Substrate in Metal-Insulator-Semiconductor Structure with Band Bending Effect," *Journal of Nanoscience and Nanotechnology*, vol. 21, no. 8, pp. 4315–4319, 2021. doi:10.1166/jnn.2021.19391

22. **H. Kim**, H. B. Yoo, J. T. Yu, J. H. Ryu, S. J. Choi, D. H. Kim, D. M. Kim, "Alternating Current-Based Technique for Separate Extraction of Parasitic Resistances in MISFETs With or Without the Body Contact," *IEEE Electron Device Letters*, vol. 41, no. 10, pp. 1528–1531, 2020. doi:10.1109/LED.2020.3020405 †

23. H. B. Yoo, J. Yu, **H. Kim**, J. H. Ryu, S. J. Choi, D. H. Kim, D. M. Kim, "Deep depletion capacitance–voltage technique for spatial distribution of traps across the substrate in MOS structures," *Solid-State Electronics*, vol. 173, pp. 107905, 2020. doi:10.1016/j.sse.2020.107905

Conference Papers (indexed)

1. H. Yang, **H. Kim**, C. Han, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Quantitative Analysis of Endurance-Dependent Charge Trapping Dynamics in Ferroelectric Field-Effect Transistors with Temperature Effects," *2025 Silicon Nanoelectronics Workshop (SNW)*, pp. 74–75, 2025. doi:10.23919/SNW65111.2025.11097276

2. H. B. Yoo, **H. Kim**, Y. Lee, J. H. Ryu, J. Y. Park, H. J. Yang, J. H. Bae, D. H. Kim, S. J. Choi, D. M. Kim, "Characterization Technique for Interface Traps in Si Nanosheet GAA MOSFETs through Subthreshold I-V Characteristics," *2022 IEEE 22nd International Conference on Nanotechnology (NANO)*, pp. 116–119, 2022. doi:10.1109/NANO54668.2022.9928778 †

Preprints

1. D. M. Kim, J. H. Ryu, H. B. Yoo, **H. Kim**, J. Y. Park, S. H. Han, J. H. Bae, S. J. Choi, D. H. Kim, "Capacitance-Based Extraction Technique for the Spatial and Energy Distributions of Traps in Si Mosfets," *SSRN*, 2023. doi:10.2139/ssrn.4377528

† Co-first author.

Research Grants and Projects

2024.09 – 2025.08	**Principal Investigator** — National Research Foundation of Korea Maximizing Energy Efficiency and Speed: Modeling and Optimization of Transient Response Characteristics in AOS-based Ferroelectric Memory Devices
2024.03 – 2024.12	**Graduate Researcher** — National Research Foundation of Korea Development of Application-customized Computing Be-spoke Device for High-density and Efficient PIM
2022.06 – 2023.05	**Graduate Researcher** — Industry–University Cooperation Research, Alsemy Co. Semiconductor device modeling using artificial neural networks
2020 – 2024	**Graduate Researcher & Staff** — National Research Foundation of Korea Four-Step Brain Korea (BK) 21 Project
2020 – 2022	**Graduate Researcher** — National Research Foundation of Korea Hybrid Device-Based Circadian ICT Research Center

Awards and Honors

2025	**Excellent Talent Award**, Kookmin University (Ph.D. conferral, Fall 2025)
2023	**Excellent Poster Award on Site**, The 30th Korean Conference on Semiconductors
2022	**Excellent Talent Award**, Kookmin University (M.S. conferral, Spring 2022)

Teaching Experience

2025	**Lecturer** — Intelligent Semiconductor Devices, Kookmin University (junior level, 3 credits; reference: Neamen, *Semiconductor Physics and Devices*)
2024	**Session Instructor** — Winter job training: Measurement of semiconductor devices and extraction of characteristic parameters, Kookmin University (senior)
2023	**Session Instructor** — Summer job training: Measurement of semiconductor devices and extraction of characteristic parameters, Kookmin University (senior)
2023	**Teaching Assistant** — Basic Electronic Experiment, Kookmin University (sophomore)
2022 – 2023	**Teaching Assistant** — Electrical Engineering Capstone Design I, Kookmin University (senior; 2022, 2023)
2020 – 2022	**Teaching Assistant** — Electrical Engineering Capstone Design II, Kookmin University (senior; 2020, 2021, 2022)
2021	**Session Instructor** — Summer job training: Semiconductor Device TCAD Simulation Training, Kookmin University (junior–senior)
2019	**Undergraduate Mentoring** — Semiconductor Engineering; Electromagnetics, Kookmin University (sophomore–junior)

Selected Conference Presentations

- **H. Kim**, S. H. Han, H. Jeong, H. Yang, S.-J. Choi, D. H. Kim, J.-H. Bae, and D. M. Kim, "Photonic I–V-based technique for subgap density of states extraction in AOS TFTs," *Int. Conf. on Electronics, Information, and Communication (ICEIC)*, Osaka, Japan, Jan. 2025.
- **H. Kim**, J. Park, J. Im, H. Kim, Y. J. Yoon, Y.-M. Kim, K. Park, S. Y. Woo, and J.-H. Bae, "Impact of Proton Irradiation on SiNx and Si–SiO_2 Interfaces in FLASH Memory," *2024 IEEE Silicon Nanoelectronics Workshop (SNW)*, Honolulu, HI, USA, 2024, pp. 1–2.
- J. Y. Park, **H. Kim**, H. B. Yoo, J. H. Ryu, J.-H. Bae, S.-J. Choi, D. H. Kim, and D. M. Kim, "Comprehensive DC Technique for Extraction of Threshold Voltage, Parasitic Resistances, and Mobility Enhancement Parameter in AOS TFTs," *The 28th Korean Conference on Semiconductors*, Jan. 2021. †
- H. B. Yoo, **H. Kim**, J. Yu, Y. J. Park, S.-J. Choi, D. H. Kim, and D. M. Kim, "Alternating Current-based Open Drain Method for Separate Extraction of Source and Drain Resistances in MOSFETs," *The 27th Korean Conference on Semiconductors*, Feb. 2020. † (presenter)

A further 17 conference abstracts (Korean Conference on Semiconductors, IEIE Summer Annual Conference) are omitted for length and are available on request.

Technical Skills

Measurement. Agilent 4156C / B1500A and Keithley 4200-SCS semiconductor parameter analyzers; HP 4284A and 4294A LCR meters; Agilent E5250A switching matrix; SUMMIT-9200 probe station; MS TECH MST-1000B hot chuck controller; laser diode sources (OZ-Series) and PM100D optical power meter; RIGOL DG5102 function generator; DPO3014 oscilloscope; Keysight 35670A dynamic signal analyzer; SR570 low-noise current preamplifier; CX3322A device current waveform analyzer; WITec alpha Raman/PL spectroscopy.

Simulation and computation. Synopsys ISE TCAD, Silvaco TCAD, OrCAD PSpice Designer, PSIM, P–CAD, MATLAB, Python.

Publication list generated from a DOI-verified source of truth (`achievements/publications.bib`); every entry was cross-checked against its Crossref record. Last updated: August 2026.